



Original article

Structural network disruption after optic neuritis in Myelin oligodendrocyte glycoprotein antibody associated disease



Suho Ro ^{a,b,1}, Daegyeom Kim ^{c,d,1}, Eun Bin Cho ^f, Yeon Hak Chung ^{g,h}, Hyunjin Ju ^{g,i},
Kyung-Ah Park ^j, Sung Tae Kim ^k, Cheol E. Han ^{c,e,**}, Ju-Hong Min ^{g,l,m,*}

^a Department of Neurology, Kangbuk Samsung Hospital, Sungkyunkwan University School of Medicine, Seoul, South Korea

^b Department of Neurology, Graduate School of Medicine, Sungkyunkwan University, Seoul, South Korea

^c Department of Electronics and Information Engineering, Korea University, Sejong, South Korea

^d Smart City Intelligent Healthcare Research Center, Korea University, Sejong, South Korea

^e Division of Smart Energy Convergence Engineering, Korea University, Sejong, South Korea

^f Department of Neurology, Gyeongsang National University Changwon Hospital, Gyeongsang National University, Jinju, South Korea

^g Department of Neurology, Samsung Medical Center, Sungkyunkwan University School of Medicine, Seoul, South Korea

^h Department of Neurology, Korea University Guro hospital, Seoul, South Korea

ⁱ Department of Neurology, Soonchunhyang university Hospital, Seoul, South Korea

^j Department of Ophthalmology, Samsung Medical Center, Seoul, South Korea

^k Department of Radiology, Samsung Medical Center, Sungkyunkwan University School of Medicine, Seoul, South Korea

^l Neuroscience Center, Samsung Medical Center, Seoul, South Korea

^m Department of Health Sciences and Technology, Samsung Advanced Institute for Health Sciences & Technology (SAIHST), Sungkyunkwan University, Seoul, South Korea

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ABSTRACT

Background: Optic neuritis is a common manifestation of myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD), which displays distinctive features that differ from other demyelinating diseases. **Objectives:** In this study, we investigated the alterations in white matter (WM) connectivity post-optic neuritis in patients with MOGAD.

Methods: We enrolled patients with MOGAD presenting with optic neuritis (MOGAD-ON) ($N = 18$) and healthy controls (HC) ($N = 35$). Structural connections between any pair of 90 cortical and subcortical regions were established using diffusion tensor imaging and graph theory. Network-based statistics were employed to identify disrupted patterns in WM networks. Additionally, we investigated the association between WM network integrity and visual acuity.

Results: Analysis of WM networks revealed that MOGAD-ON patients exhibited reduced total strength, global efficiency, and local efficiency compared to HCs (all, $p < 0.05$). At the nodal level, MOGAD-ON patients showed more regions with altered network topologies than HCs, some of which were associated with visual acuity and visual functional scores. NBS identified the most discriminative connectivity changes in the left hippocampus, thalamus, inferior temporal gyrus, fusiform gyrus, and middle temporal occipital gyrus.

Conclusion: This study indicates that MOGAD-ON patients experience significant disruptions in WM networks, particularly in the left temporal and occipital lobes.

* Corresponding author at: Department of Neurology, Sungkyunkwan University School of Medicine, Neuroscience Center, Samsung Medical Center, Department of Health Sciences and Technology, Samsung Advanced Institute for Health Sciences & Technology (SAIHST), Sungkyunkwan University, 50 Irwon-dong, Gangnam-gu, Seoul 135-710, South Korea.

** Corresponding author at: Department of Electronics and Information Engineering, Korea University, Sejong, Division of Smart Energy Convergence Engineering, Korea University, 2511 Sejong-ro, Sejong, 30019, South Korea.

E-mail addresses: cheolhan@korea.ac.kr (C.E. Han), juhongm@skku.edu (J.-H. Min).

¹ Both authors equally contributed to this work as first authors.

1. Introduction

Optic neuritis (ON) is a common manifestation of CNS inflammatory demyelinating diseases, including multiple sclerosis (MS), neuromyelitis optica spectrum disorder (NMOSD), and myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD). Recent advancements in diagnostic criteria have underscored distinct pathophysiological mechanisms in MOGAD compared to MS and NMOSD, thereby emphasizing the necessity for disease-specific diagnostic and therapeutic approaches (Banwell et al., 2023).

MRI characteristics are different in patients experiencing an ON attack. MOG-ON and AQP4-ON often exhibit more severe T2 hyperintensity, unlike MS-ON, where mild or no hyperintensity is more common. Furthermore, brain T2-hyperintense lesions are absent in 47–68 % of MOGAD patients at onset with ON or myelitis, whereas these lesions appear in the majority of MS patients (84–92 %) and a significant portion of NMOSD patients (51 %) at onset (Banwell et al., 2023; Cobo-Calvo et al., 2018; Huh et al., 2014; Ramanathan et al., 2016). Although these imaging features are well-documented, quantitative MRI studies have revealed subtle abnormalities in both lesional and non-lesional CNS tissues in MOGAD, suggesting microstructural damage beyond visible white matter (WM) lesions (Brier et al., 2023). This damage often disrupts functional network connectivity, increasingly recognized as linked to neurological symptoms (Friston et al., 2016). For instance, NMOSD patients, even in the absence of brain lesions, exhibit disrupted network connectivity, indicating the potential for diffuse network dysfunction in inflammatory demyelinating conditions (Liu et al., 2020). Similarly, functional MRI studies on clinically isolated syndrome (CIS) and NMOSD have shown alterations in visual network density, further emphasizing the significance of network-level disruptions (Backner et al., 2020).

Despite advancements in understanding MOGAD, the impact of this disease on brain network connectivity remains underexplored, particularly in ON patients lacking detectable brain lesions. Our study is dedicated to investigating changes in brain networks in such patients, aiming to elucidate the wider neural impacts of MOGAD and their potential clinical implications.

2. Methods

2.1. Subjects

We prospectively enrolled 18 MOGAD patients who presented with ON and attended the clinic from 2015 to 2022 as part of a prospective cohort study on CNS demyelinating diseases at Samsung Medical Center, Seoul, South Korea. Inclusion criteria included patients (1) aged over 17 years, (2) meeting the MOGAD diagnostic criteria (Banwell et al., 2023), (4) with a history of ON without symptomatic brain involvement, and (5) having available brain MRIs at remission, performed at least 3 months after the onset of ON.

We collected the exponentially converted visual acuity values and the Expanded Disability Status Scale (EDSS) scores, including the visual functional systems (FS) score (Kurtzke, 1983). Additionally, 35 healthy controls (HC) with no history of medical, neurological, or psychiatric disorders were included. This study was approved by the Institutional Review Board of Samsung Medical Center (IRB 2020-04-119) and conducted in accordance with the Declaration of Helsinki, with written informed consent obtained from all participants.

2.2. Magnetic resonance imaging acquisition

All participants underwent a 3D volumetric brain MRI scan using an Achieva 3.0-Tesla MRI scanner (Philips, Best, the Netherlands). High-resolution anatomical images were acquired using both a 3D T1-weighted sequence, and a fluid-attenuated inversion recovery (FLAIR) sequence, whose voxel size is 0.5 mm isovoxel. Whole-brain diffusion

tensor images (DTI) were acquired with voxel dimensions of $1.72 \times 1.72 \times 2 \text{ mm}^3$, a b-factor of 600 s/mm^2 , and 45 diffusion directions. Detailed acquisition protocols are provided in Supplementary Material S1.

2.3. Network construction

To construct the brain network, nodes were defined as individual brain regions, while edge weights represented the average fractional anisotropy (FA) values along fiber tracts interconnecting these regions. The brain regions were delineated by registering the automated anatomical labeling (AAL) atlas to each subject's DTI using SPM12. We obtained the registered AAL atlas in each subject's DTI, encompassing 78 cortical and 12 subcortical regions, excluding the cerebellar areas.

The edge weight between any two brain regions was calculated by averaging the FA values of all voxels traversed by the connecting streamlines. Whole brain tractography began by adjusting the DTI to negate any unwanted movements, aligning all volumes with gradient directions to a reference volume using the eddy tool of FSL's diffusion toolkit (v 5.0.10) (Andersson and Sotiropoulos, 2016). Within this process, the gradient directions were suitably rotated. We preprocessed the corrected DTI images and conducted tractography utilizing the diffusion toolkit in conjunction with TrackVis (Mori and Barker, 1999; R. wang et al., 2007). Employing the Fiber Assignment by Continuous Tracking algorithm with a 45-degree angular threshold, we excluded voxels with FA values below 0.2 in the seed voxels and omitted streamlines shorter than 20 mm to reduce artifacts. The average FA value of all voxels along the streamline connection paths was then computed. As a result, each subject's brain network consisted of 90 regions, where the edge weights quantify the average FA values across all possible pairs of brain regions in a 90-by-90 connectivity matrix.

2.4. Lesion volume segmentation

To explore the interhemispheric differences in lesions among MOGAD-ON patients, we quantified the lesion volumes in both hemispheres. All lesions were manually outlined on FLAIR images using MRIcron software. Subsequently, a Wilcoxon signed rank test was employed to compare the lesion volumes between the patient's left and right hemispheres.

To ensure the reliability of the manual segmentation, radiological findings were analyzed independently by two neurologists blinded to the diagnosis. Prior to the study, both neurologists underwent a standardized training session on the lesion delineation protocol to minimize inter-rater variability, which could not be statistically quantified due to the small number of lesions. This training included reviewing a consensus set of challenging cases and establishing clear guidelines for lesion boundary identification on T2-weighted FLAIR images. Care was taken to exclude surrounding cerebrospinal fluid (CSF) or partial volume effects, which could be frequently misclassified as true lesions in automated segmentation tool.

2.5. Network measures

The Brain Connectivity Toolbox facilitates the analysis of various network measures that quantify the global and nodal characteristics of networks (Rubinov and Sporns, 2010). We measured degree, strength, density, regional efficiency, global efficiency, and local efficiency through a MATLAB (R2022a, MathWorks, Natick, USA) implementation.

The nodal degree and nodal strength characterize the direct influence of a node on its neighbors. The former is the number of neighboring nodes connected to a node, while the latter is the sum of the edge weights connected to the node. At the network level, the edge density and total strength correspond to nodal degree and nodal strength, respectively. Edge density represents the proportion of existing connections to all possible connections. Whereas, the total strength is the

sum of all edge weights across the network. Low values of these measures may indicate WM deterioration due to neurodegeneration, which disrupts the integrity of WM tracts, weakens the edge weights of the affected connections, and causes edge disconnections. Consequently, these connectivity changes may influence network efficiency, where efficiency measures reflect the ability of a network to transmit information. Regional efficiency quantifies how efficiently a specific node communicates with all the other nodes. Its network-level counterpart, global efficiency measures the overall efficiency of information transfer. Local efficiency assesses the efficiency of information transfer within a node's neighborhood. Local efficiency at the network level represents the overall resilience of local subnetworks. Formal descriptions of these measures are provided in Supplementary Material S2.

2.6. Network based statistics (NBS)

To identify a subnetwork characterized by significant differences in WM connections between the MOGAD and HC groups, NBS was employed, utilizing an analysis of covariance (ANCOVA) while controlling for age and sex effects. The threshold was set when stable clusters emerged across multiple iterations. The selected threshold was 2.3, and 10,000 permutations were conducted (For further details, see Supplementary Material S3). Furthermore, the hub nodes of the subnetwork, representing brain regions most impacted by the identified abnormal structural connectivity, were defined when their degree exceeded the mean plus two standard deviations of all regions connected by the edges identified by NBS.

2.7. Statistical analysis

We compared both global and nodal network measures among two groups using permutation-based ANCOVA (Cho et al., 2018), adjusting for the effects of age and sex with 10,000 permutations. Multiple comparisons for nodal measures were corrected using Bonferroni across 90 nodes. We used our in-house codes for permutation-based ANCOVA under MATLAB (R2022a, MathWorks, Natick, USA). For detailed these procedures, see Supplementary Material S4.

We investigated the relationship between the network measures from identified hub nodes against clinical symptoms incorporating visual acuity and visual FS scores through Spearman's correlation, focusing on nodes that demonstrated statistically significant group differences between MOG and HC.

3. Results

3.1. Demographics and clinical features

Demographics and clinical features of MOGAD-ON patients and HCs are summarized in Table 1. Age and sex ratios were comparable between the two groups. The average disease duration was 1060.4 ± 969.6 days, with a total number of attacks reported as 2 (median, IQR 2–3). The median EDSS score was 1.0 (IQR 1.0–2.0), and the median visual functional score was 1 (IQR 0–2). Patients have been treated with oral corticosteroid ($N = 2$), azathioprine ($N = 8$), and mycophenolate ($N = 2$).

Radiologically, 9 of the 18 patients (50 %) exhibited nonspecific white matter lesions on brain MRI. These lesions were primarily located in the periventricular or deep subcortical white matter. One patient had an additional small lesion in the basal ganglia and internal capsule. None of these lesions corresponded to typical MOGAD-related brain involvement and they were symmetrically distributed without hemispheric predominance ($p = 0.2574$, Table S1) (Banwell et al., 2023). We note that lesions among these nine patients were spatially scattered, resulting in few voxel-wise overlaps across subjects. The remaining nine patients had no detectable FLAIR lesions.

Structured neurological history taking and physical examination by

Table 1

Demographic and clinical characteristics of MOGAD-ON and HC groups.

	MOGAD-ON (n = 18)	HC (n = 35)	P-values
Sex, N of females (%)	10 (55.56)	21 (60.00)	0.7558
Age, mean $\pm$ SD in years	43.61 $\pm$ 15.61	37.71 $\pm$ 12.61	0.1549
Right-handedness	17(94.44)	NA	
Disease duration, mean $\pm$ SD in days	1060.44 $\pm$ 969.60	NA	
Visual acuity, median (IQR) scaled			
Right	0.0 [0.0,0.0] ¹	NA	
Left	0.0 [0.0,0.2] ¹	NA	
Visual functional score, median (IQR)	1 [0,2] ¹	NA	
Expanded Disability Status Scale (EDSS) score, median (IQR)	1.0[1.0,2.0] ¹	NA	
Attack numbers, median (IQR)	2 [2,3] ¹	NA	
Medication, N (%)	12 (66.67)	NA	
Oral corticosteroid	2 (11.11)		
Azathioprine	8 (44.44)		
Mycophenolate	2 (11.11)		
Immunoglobulin	0		

MOGAD-ON, myelin oligodendrocyte glycoprotein antibody-associated disease patients who presented with optic neuritis; HC, healthy controls; ¹median [IQR]; NA, not applicable.

neurologists were performed in all patients, none of whom reported or exhibited neurological symptoms other than optic neuritis, and any prior history of neurological deficits. The observed lesions were considered clinically silent in all cases.

3.2. Disruption of brain network topology in MOGAD

3.2.1. Global measures

Overall, the MOGAD-ON group demonstrated lower total strength, global efficiency, and local efficiency compared to the HC group ($p = 0.0024$, $p = 0.0009$, and $p = 0.0064$, respectively), while edge density showed no significant differences between the groups (Table 2).

3.2.2. Nodal measures

The results of nodal measures are summarized in Table S2 and Fig. 1. The MOGAD-ON group exhibited lower nodal strength in the left parahippocampal gyrus, left fusiform gyrus, left superior parietal gyrus, left inferior temporal gyrus, and left amygdala. Differences in local and regional efficiency were widely distributed across hemispheres, predominantly in the left temporal and occipital cortex.

3.2.3. Correlation analysis with visual acuity

We identified several regions where network measures correlated with visual acuity or FS score in 17 MOGAD-ON patients, excluding one outlier with exceptionally poor visual acuity (VA=0.15/0.3).

Table 2

Global network topological measures in MOGAD-ON and HC groups.

Network Measures	MOGAD-ON	HC	Statistics ¹
Total Strength	290.4086 $\pm$ 21.2603	317.2823 $\pm$ 27.6407	F = 10.6396, p = 0.0024
Edge Density	0.2019 $\pm$ 0.0131	0.2041 $\pm$ 0.0156	F = 0.1695, p = 0.6756
Global Efficiency	0.2292 $\pm$ 0.0177	0.2478 $\pm$ 0.0149	F = 13.1314, p = 0.0009
Local Efficiency	0.3119 $\pm$ 0.0311	0.3363 $\pm$ 0.0237	F = 8.2715, p = 0.0064

MOGAD-ON, myelin oligodendrocyte glycoprotein antibody-associated disease patients who presented with optic neuritis; HC, healthy controls; Data are presented as mean $\pm$ standard deviation.¹F values and p values are derived from a permutation-based ANCOVA controlling for age and sex between two groups.

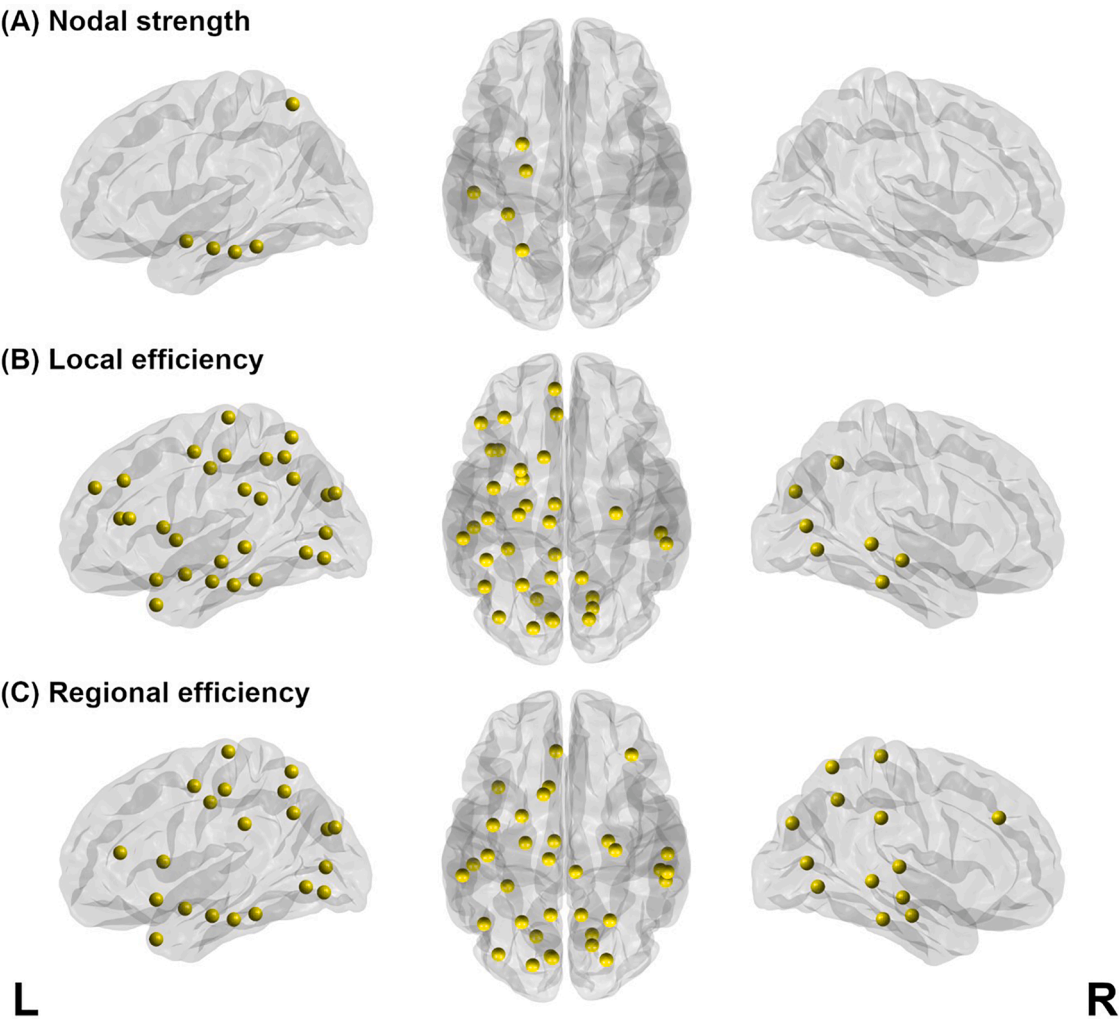


Fig 1. Topological differences between MOGAD-ON and HC groups. MOGAD-ON, an abbreviation for myelin oligodendrocyte glycoprotein antibody-associated disease patients who presented with optic neuritis, demonstrated lower nodal strength, local efficiency, and regional efficiency. Nodes that exhibited statistically significant differences were highlighted with yellow dots. Among the three figures presented, the leftmost (labeled 'L') depicts the left hemisphere, the rightmost (labeled 'R') the right hemisphere, and the middle figure illustrates both hemispheres from a superior view. The disparities in local and regional efficiency between the two groups were pronounced across hemispheres, particularly in the left temporal and occipital cortex.

Most network measures were negatively associated with visual acuity or FS score, whereas nodal strength and regional efficiency of the left superior parietal gyrus showed positive correlations with left visual acuity (scaled) ($\rho=0.5173$ and $\rho=0.5071$; both $p < 0.05$). These findings are detailed in [Table 3](#).

Table 3
Correlation between network measures, visual acuity (scaled), and FS score (EDSS, scaled) in MOGAD-ON patients.

Nodal Measure	VA_L Scaled			VA_R Scaled			FS Scaled		
	Node	r-value	p-value	Node	r-value	p-value	Node	r-value	p-value
Nodal Strength	Amygdala_L	−0.6714	0.0032						
	Parietal_Sup_L	0.5173	0.0335						
Local Efficiency	Hippocampus_R	−0.4854	0.0482	Precentral_L	−0.5700	0.0169			
				Frontal_Mid_L	−0.5942	0.0119			
				Postcentral_L	−0.4999	0.0410			
				Angular_L	−0.5107	0.0362			
				Putamen_L	−0.5484	0.0226			
				Temporal_Mid_R	−0.5295	0.0288			
Regional efficiency	Frontal_Mid_R	−0.5555	0.0206	Precentral_L	−0.7020	0.0017	Frontal_Mid_R	−0.4881	0.0468
	Hippocampus_R	−0.5313	0.0282	Olfactory_L	−0.5053	0.0386	Temporal_Mid_R	−0.5342	0.0272
	Amygdala_L	−0.4880	0.0469	Postcentral_L	−0.7168	0.0012			
	Parietal_Sup_L	0.5071	0.0378	Angular_L	−0.6319	0.0065			
	Temporal_Mid_R	−0.5389	0.0256	Caudate_L	−0.5201	0.0323			

All correlations were conducted using Spearman’s method. FS, visual functional score; EDSS, Expanded Disability Status Scale; MOGAD-ON, myelin oligodendrocyte glycoprotein antibody-associated disease patients who presented with optic neuritis; VA, visual acuity; L, left; R, right; Sup, superior; Mid, middle; Inf, inferior.

3.3. Network-based statistics

In NBS analysis, network disruptions in MOGAD-ON patients, when compared to HCs, comprised 24 edges, 22 nodes, and 6 hub nodes ($p = 0.006$) (Table S3 and Fig. 2). These disturbances were detected primarily in the left deep grey matter and left temporo-occipital lobes, while also affecting some interhemispheric connections. The affected regions included six hub nodes: left hippocampus, left thalamus, left inferior temporal gyrus, left fusiform gyrus, left lingual gyrus, and left middle occipital gyrus.

4. Discussion

We demonstrate that MOGAD patients with ON exhibit significant alterations in WM connectivity, especially within the visual network and its associated regions. The most distinct disruptions were identified in the left temporo-occipital lobes and left thalamus. Notably, visual acuity was strongly correlated with nodal network measures in regions not traditionally viewed as part of the core visual pathway, such as the left amygdala, precentral gyrus, postcentral gyrus, and angular gyrus. These findings underscore the complex interplay between visual function and broader network connectivity.

Impaired brain connectivity in MOGAD patients was confirmed by alterations in global measures. The brain regions most significantly impacted, indicated by 6 hub nodes. The visual pathway is composed of the lateral geniculate nucleus, the optic radiation traversing the temporal and parietal lobes, and the visual cortex, including the occipital lobe (Rodrigues et al., 2022). The functional roles of these regions underscore their contributions to the integration of visual perception, memory, and spatial processing within the visual network. (Miyashita, 1993; Weiner and Zilles, 2016). The hippocampus plays a role in the visual discrimination of complex spatial scenes (Lee et al., 2012). The lingual and middle occipital gyri are crucial for processing static and dynamic visual stimuli, respectively (Grill-Spector et al., 2001; McCarthy et al., 1999). Previously, patients diagnosed with NMOSD or CIS who presented with ON demonstrated decreased connectivity within the visual network, particularly in dorsal-lateral regions including the middle temporal region, superior temporal area, lateral occipital cortex, and visual areas 2 and 3 (Backner et al., 2020). Recent studies in MOGAD have shown that ON can lead to trans-synaptic degeneration along the visual pathway, including structural atrophy in the lateral geniculate nucleus and occipital cortex, which is in line with our results, more pronounced altered connectivity in the visual pathway areas (Bollo et al., 2024).

Most regions showed a negative association between nodal network measures and visual function, except superior parietal gyrus. The amygdala contributes to visual awareness by receiving visual input, assigning emotional and motivational significance, and relaying this information back to visual areas. (Duncan and Barrett, 2007; Iwai et al., 1990). The middle temporal gyrus houses a motion-sensitive visual association area known as visual cortex 5 (Lee et al., 2012; Thomas et al., 2013). The precentral gyrus, middle frontal gyrus, angular gyrus, and caudate participate in voluntary eye movement (Pouget, 2015). The superior parietal gyrus is integral to sensation discrimination, visual function, and spatial awareness within the dorsal visual pathway. (van Polanen and Davare, 2015) Conversely, the postcentral gyrus plays a supplementary role in visuospatial attention. (Balslev et al., 2013) Furthermore, neurons in the caudal-ventral part of the putamen process visual information, encompassing spatial and object characteristics (Kunimatsu et al., 2019). The olfactory gyrus directly relays information to the limbic system and connects with the amygdala and hippocampus (Sullivan et al., 2015). Significantly, strong correlations between nodal measures and visual acuity were observed in the amygdala, precentral, postcentral, and angular gyri, which are not the primary areas consisting of the visual pathway. This indicates that the visual function in MOGAD patients with ON is associated with altered connectivity in visual network-associated areas responsible for visual awareness, perception, and re-processing. Moreover, visual function correlates more with areas that are not the main structures of the visual system. A functional neuroimaging study reported a significant network disruption in extravisual regions as well as visual cortices in MS patients with ON, suggesting that acute inflammation in ON may induce diffuse connectivity changes (Stampanoni Bassi et al., 2017). Other studies have also demonstrated altered functional connectivity in patients with ON, involving regions such as the middle frontal gyrus, middle temporal gyrus, amygdala, angular gyrus, and superior parietal gyrus, findings that are consistent with our own (Huang et al., 2018; Song et al., 2021). Additionally, NMOSD patients with ON exhibited neural network dysfunction in the left frontal superior orbital gyrus, left angular gyrus, and right parietal superior gyrus (Wei et al., 2022). Intriguingly, increased network connectivity in the left superior parietal gyrus was associated with better visual function in our MOGAD patients, which may suggest compensatory neural adaptations, as observed in previous studies (Stampanoni Bassi et al., 2017).

Interestingly, network disruption was predominantly localized to the left hemisphere, consistent with established anatomical asymmetries and functional lateralization. Prior studies have demonstrated leftward asymmetries in areas related to language, visual processing, and sensory

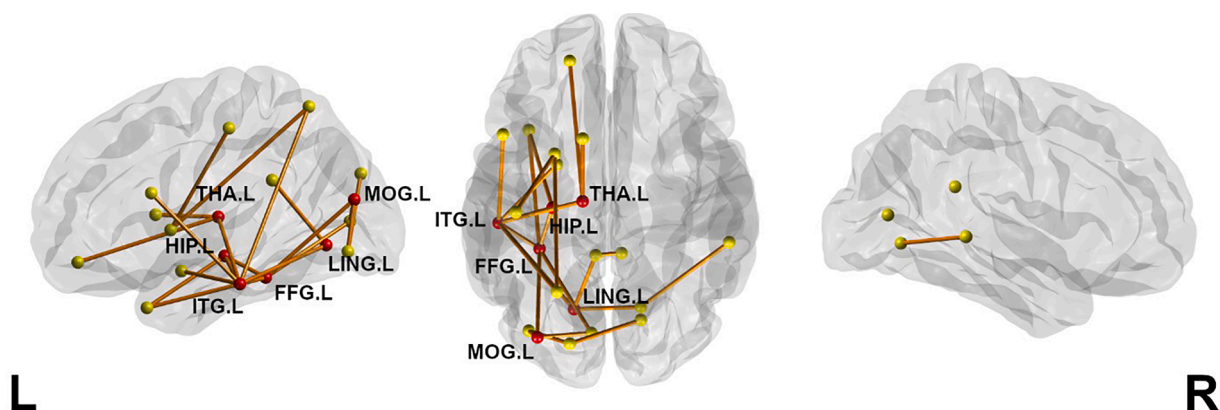


Fig 2. The abnormal subnetwork in MOG-ON. In the MOGAD-ON group, we identified a subnetwork with reduced connectivity compared to the HC group using network-based statistics (NBS), characterized by orange lines denoting abnormal connections. Brain regions associated with these abnormal connections were indicated by yellow dots, and hub nodes were represented by red dots. Among the three figures presented, the leftmost (labeled 'L') depicts the left hemisphere, the rightmost (labeled 'R') the right hemisphere, and the middle figure illustrates both hemispheres from a superior view. MOGAD-ON denotes myelin oligodendrocyte glycoprotein antibody-associated disease patients with optic neuritis; ITP.L is the left inferior temporal gyrus; FFG.L refers to the left fusiform gyrus; LING.L stands for the left lingual gyrus; MOG.L is the left middle occipital gyrus; HIP.L designates the left hippocampus, and THA.L is the left thalamus.

functions (Shu et al., 2015). These findings are consistent with our observations, suggesting that topological asymmetries in the anatomical network may reflect functional lateralization of the human brain. In MS, similar left-hemisphere predominance has been reported. A notable correlation has also been observed between EDSS scores and lesion location in this region. (Charil et al., 2003; Shahbodaghy et al., 2023). However, in our study, asymptomatic WM changes were minimal. This echoes our findings, further underscoring the role of left-hemisphere structural and functional vulnerabilities in brain network disruption.

This study has several limitations. First, it included only a small cohort of patients from a single center, which may restrict the generalizability of the results to other ethnic populations or healthcare setting and lesson statistical power for subgroup analyses, although MOGAD is a relatively rare demyelinating disorder of the CNS. Second, most of the patients exhibited good visual acuity, commonly known to improve swiftly after corticosteroid treatment in MOGAD (Banwell et al., 2023). Third, MRI data cross-sectionally collected for three months post-onset might not reflect the long-term changes in the brain network subsequent to optic neuritis. Longitudinal research with prolonged follow-ups would offer much-needed insights into chronic clinical consequences, such as the likelihood of additional relapses. Fourth, it remains whether current disease-modifying treatments as well as disease activity in MOGAD could affect trans-synaptic neuroaxonal degeneration or network disruption, which remains to be further investigated in the large cohort studies. Lastly, inherent limitations of DTI and deterministic tractography, like their inability to depict crossing fibers, might have influenced the outcomes. Considering alternative techniques, such as probabilistic tractography, for future studies might be beneficial.

5. Conclusion

This study highlights disruptions in WM connectivity within the visual network and adjacent regions in MOGAD patients with ON, who have no history of noticeable symptomatic brain attacks. Importantly, the correlations between nodal network metrics and visual acuity indicate that these disruptions in connectivity directly relate to the severity of visual impairment. Additional research involving larger cohorts and longitudinal approaches is necessary to better understand the physiological and clinical ramifications of these disruptions in MOGAD-ON.

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Data availability statement

Fully anonymized data are available after review by Samsung medical center. An application form detailing specific requirements, rationale and proposed use should be completed, followed by a data-sharing agreement. Requested data may be made available along with supporting documentation (e.g. data dictionary) on a secure server to

appropriate and approved investigators.

CRediT authorship contribution statement

Suho Ro: Writing – original draft, Validation, Investigation, Data curation, Conceptualization. **Daegyeom Kim:** Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Eun Bin Cho:** Writing – review & editing. **Yeon Hak Chung:** Writing – review & editing. **Hyunjin Ju:** Writing – review & editing. **Kyung-Ah Park:** Writing – review & editing. **Sung Tae Kim:** Writing – review & editing. **Cheol E. Han:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Ju-Hong Min:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

JH Min is funded by and has received research support from the National Research Foundation of Korea (MIST and KHIDI) and SMC Research and Development Grant. She has lectured, consulted and received honoraria from Bayer Healthcare, Merk, Biogen Idec, Sanofi, UCB, Samsung Bioepis, Mitsubishi Tanabe, Celltrion, Roche, Janssen, Astrazeneca and MedicinePharma. All other authors have no relevant financial or non-financial interests to disclose.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.msard.2025.106862](https://doi.org/10.1016/j.msard.2025.106862).

References

- Andersson, J.L.R., Sotiropoulos, S.N., 2016. An integrated approach to correction for off-resonance effects and subject movement in diffusion MR imaging. *Neuroimage* 125, 1063–1078.
- Backner, Y., et al., 2020. Cortical topological network changes following optic neuritis. *Neurol. Neuroimmunol. Neuroinflamm.* 7 (3).
- Balslev, D., Odoj, B., Karnath, H.O., 2013. Role of somatosensory cortex in visuospatial attention. *J. Neurosci.* 33 (46), 18311–18318.
- Banwell, B., et al., 2023. Diagnosis of myelin oligodendrocyte glycoprotein antibody-associated disease: international MOGAD Panel proposed criteria. *Lancet Neurol.* 22 (3), 268–282.
- Bollo, L., et al., 2024. Trans-synaptic degeneration in the visual pathway in patients with myelin oligodendrocyte glycoprotein antibody-associated disease. *Neurology.* 102 (7), e209156.
- Brier, M.R., et al., 2023. Quantitative MRI identifies lesional and non-lesional abnormalities in MOGAD. *Mult. Scler. Relat. Disord.* 73, 104659.
- Charil, A., et al., 2003. Statistical mapping analysis of lesion location and neurological disability in multiple sclerosis: application to 452 patient data sets. *Neuroimage* 19 (3), 532–544.
- Cho, E.B., et al., 2018. White matter network disruption and cognitive dysfunction in neuromyelitis optica spectrum disorder. *Front. Neurol.* 9, 1104.
- Cobo-Calvo, A., et al., 2018. Clinical spectrum and prognostic value of CNS MOG autoimmunity in adults: the MOGADOR study. *Neurology* 90 (21), e1858–e1869.
- Duncan, S., Barrett, L.F., 2007. Affect is a form of cognition: a neurobiological analysis. *Cogn. Emot.* 21 (6), 1184–1211.
- Friston, K., et al., 2016. The dysconnection hypothesis (2016). *Schizophr. Res.* 176 (2–3), 83–94.
- Grill-Spector, K., Kourtzi, Z., Kanwisher, N., 2001. The lateral occipital complex and its role in object recognition. *Vis. Res.* 41 (10–11), 1409–1422.
- Huang, J., et al., 2018. Altered brain structure and functional connectivity of primary visual cortex in optic neuritis. *Front. Hum. Neurosci.* 12, 473.
- Huh, S.Y., et al., 2014. The usefulness of brain MRI at onset in the differentiation of multiple sclerosis and seropositive neuromyelitis optica spectrum disorders. *Mult. Scler.* 20 (6), 695–704.
- Iwai, E., et al., 1990. A role of amygdala in visual perception and cognition in macaque monkeys (*Macaca fuscata* and *Macaca mulatta*). *Tohoku J. Exp. Med.* 161 Suppl, 95–120.
- Kunimatsu, J., Maeda, K., Hikosaka, O., 2019. The caudal part of putamen represents the historical object value information. *J. Neurosci.* 39 (9), 1709–1719.

- Kurtzke, J.F., 1983. Rating neurologic impairment in multiple sclerosis: an expanded disability status scale (EDSS). *Neurology* 33 (11), 1444–1452.
- Lee, A.C., Yeung, L.K., Barense, M.D., 2012. The hippocampus and visual perception. *Front. Hum. Neurosci.* 6, 91.
- Liu, Y., et al., 2020. Abnormal baseline brain activity in neuromyelitis optica patients without brain lesion detected by resting-state functional magnetic resonance imaging. *Neuropsychiatr. Dis. Treat.* 16, 71–79.
- McCarthy, G., et al., 1999. Electrophysiological studies of human face perception. II: response properties of face-specific potentials generated in occipitotemporal cortex. *Cereb. Cortex* 9 (5), 431–444.
- Miyashita, Y., 1993. Inferior temporal cortex: where visual perception meets memory. *Annu. Rev. Neurosci.* 16, 245–263.
- Mori, S., Barker, P.B., 1999. Diffusion magnetic resonance imaging: its principle and applications. *Anat. Rec.* 257 (3), 102–109.
- Pouget, P., 2015. The cortex is in overall control of 'voluntary' eye movement. *Eye (L)* 29 (2), 241–245.
- R.wang, T.B., Sorensen, A.G., Wedeen, V.J., 2007. Diffusion toolkit: a software package for diffusion imaging data processing and tractography. *Proc. Intl. Soc. Mag. Reson. Med.* 1.
- Ramanathan, S., et al., 2016. Radiological differentiation of optic neuritis with myelin oligodendrocyte glycoprotein antibodies, aquaporin-4 antibodies, and multiple sclerosis. *Mult. Scler.* 22 (4), 470–482.
- Rodrigues, E.M., et al., 2022. Anatomy of the optic radiations from the white matter fiber dissection perspective: a literature review applied to practical anatomical dissection. *Surg. Neurol. Int.* 13, 309.
- Rubinov, M., Sporns, O., 2010. Complex network measures of brain connectivity: uses and interpretations. *Neuroimage* 52 (3), 1059–1069.
- Shahbodaghy, F., et al., 2023. Symmetry differences of structural connectivity in multiple sclerosis and healthy state. *Brain Res. Bull.* 205, 110816.
- Shu, N., et al., 2015. Hemispheric asymmetry of human brain anatomical network revealed by diffusion tensor tractography. *Biomed. Res. Int.* 2015, 908917.
- Song, K., et al., 2021. Altered small-world functional network topology in patients with optic neuritis: a resting-state fMRI study. *Dis. Markers* 2021, 9948751.
- Stampanoni Bassi, M., et al., 2017. Remodeling functional connectivity in multiple sclerosis: a challenging therapeutic approach. *Front. Neurosci.* 11, 710.
- Sullivan, R.M., et al., 2015. Olfactory memory networks: from emotional learning to social behaviors. *Front. Behav. Neurosci.* 9, 36.
- Thomas P. Naidich, E.A.N., Pedro Pasik, *Imaging of the brain. Cerebral Cortex Vol. Chapter 10, Cerebral Cortex.* 2013. 154–173.
- van Polanen, V., Davare, M., 2015. Interactions between dorsal and ventral streams for controlling skilled grasp. *Neuropsychologia* 79 (Pt B), 186–191.
- Wei, R., et al., 2022. Irregular degree centrality in neuromyelitis optica spectrum disorder patients with optic neuritis: a resting-state functional magnetic resonance imaging study. *Mult. Scler. Relat. Disord.* 59, 103542.
- Weiner, K.S., Zilles, K., 2016. The anatomical and functional specialization of the fusiform gyrus. *Neuropsychologia* 83, 48–62.